Seat Majorities without Vote Majorities: Coalition Inversions in Brazil's Chamber of Deputies

Abstract

Proportional representation is typically evaluated by asking whether parties receive seats in proportion to votes. In fragmented legislatures, however, the politically relevant unit is often not a party but a coalition. This article examines coalition-level election inversions in Brazil's Chamber of Deputies. A coalition inversion occurs when a set of parties controls a legislative seat majority while receiving less than half of the national federal-deputy vote. Using party vote totals, elected-seat totals, cabinet-period coalition data, and party ideology classifications for the 2014, 2018, and 2022 federal-deputy elections, I identify inversions in two analytically distinct objects: observed cabinet coalitions and ideological intervals defined over the election-year party set. Five observed cabinet periods satisfy the inversion criterion: two during the mandate that followed the 2014 election, two during the mandate that followed the 2018 election, and one during the mandate that followed the 2022 election. The two 2018 cases each combine 47.25 percent of the vote with exactly 257 seats. An all-interval ideological search finds minimal connected interval inversions in 2014 and 2022, but none in 2018, including a compact 2022 PP–PL interval with 45.35 percent of the vote and 258 of 513 seats. The results show that Brazil's federally allocated and highly fragmented proportional system can generate coalitional inversions: party sets with less than half of the national vote can nevertheless command legislative majorities.

1 Introduction

Electoral systems convert votes into seats, and those seats define the legislative conditions under which governments are formed. Proportional representation is usually defended on the grounds that this conversion preserves the distribution of electoral support more faithfully than plurality or majoritarian systems. That claim, however, is often evaluated at the party level. In fragmented party systems, political authority is frequently organized at the coalition level. Governments are formed by party sets, and those party sets may obtain legislative majorities without receiving national vote majorities.

This raises a precise question: can party sets that fall short of a national vote majority nevertheless command a seat majority in the Chamber of Deputies? The concern is not exact proportionality. In any finite legislature, vote shares must be converted into whole seats, so some deviation from quota is unavoidable. Rather, what matters is whether those deviations accumulate across parties in ways that change the majority status of coalitions.

Most evaluations of proportionality stop before the point at which parliamentary authority is exercised. A party with a positive seat differential has more seats than its national vote share would imply, while another with a negative differential has fewer. These are familiar objects in the study of apportionment and proportional representation. But coalition formation aggregates these differentials. A set of parties may accumulate enough positive differentials to cross the seat-majority threshold before crossing the vote-majority threshold. When this happens, the legislature contains a majority that is not backed by a corresponding national vote majority for the same set of parties.

I use the Brazilian Chamber of Deputies to study this problem. The Chamber has 513 seats, so an absolute majority requires 257 seats. Federal deputies are elected in state-level districts under open-list proportional representation, but this article evaluates party votes and party seats after national aggregation. That is the level at which coalition claims about the Chamber are usually made: party groups are treated as national legislative blocs, even though their seats are produced through district-level contests. The design therefore isolates a specific transformation. It asks how federal-deputy votes, aggregated nationally by party, are translated into national legislative coalitions.

The analysis proceeds in two stages. First, I examine observed cabinet-period coalitions. These are the main empirical objects because they are realized party sets rather than merely possible coalitions. For each cabinet period in the mandates following the 2014, 2018, and 2022 elections[1], I translate cabinet parties into the party labels that existed at the corresponding election and calculate their vote share, seat share, quota, and seat differential[2].

Cabinet periods are defined using the party labels recorded in the cabinet data at the time of each appointment. Before vote and seat totals are calculated, those cabinet labels are translated into the party labels used in the election that produced the relevant Chamber delegation. One-to-one renames are mapped backward to the election label. Fusion cases are handled by first identifying the cabinet-period successor party and then expanding it into its election-year antecedents.

Second, I examine ideological intervals defined over the party set observed in the corresponding election. Given an ideological ordering of those parties, I enumerate every contiguous no-gap interval and identify intervals that control a seat majority without a vote majority. This second step does not replace the cabinet analysis. It asks whether inversions are also visible in the ordered ideological structure of the Chamber delegation.

The results identify five observed cabinet-period inversions. The first, 2016.2, lasted only two days, from April 14 to April 15, 2016. It is an event-window diagnostic around the impeachment rupture rather than evidence of a stable governing configuration. The second, 2017.1, is the more politically meaningful Temer-period case. After translating cabinet labels into the 2014 election-year party labels, the coalition held 266 of 513 seats while corresponding to 49.91 percent of the national federal-deputy vote. The 2018-linked periods 2021.3 and 2022.1 each have 47.25 percent of the vote and exactly 257 seats. The fifth case, 2023.1, is the initial Lula coalition after the 2022 election. It held 263 seats with 48.89 percent of the vote.

The ideological interval results reinforce the point while adding structure. In 2014, there are 528 contiguous intervals in the election-year party order, 180 of which reach a seat majority, and 8 of which are coalition inversions. Four are minimal connected interval inversions[3]. In 2018, there are 630 intervals and 172 seat-majority intervals, but no interval inversion. In 2022, there are 528 intervals, 99 seat-majority intervals, 6 inversions, and 2 minimal connected interval inversions. The most striking case is the 2022 PP–PL interval: it contains 8 parties, obtains only 45.35 percent of

[1] The 2014 election is linked to the mandate running from January 1, 2015 to December 31, 2018. This covers Dilma Rousseff's second administration until her suspension on May 12, 2016, and Michel Temer's administration from May 12, 2016 through the end of 2018. The 2018 election is linked to Jair Bolsonaro's mandate, from January 1, 2019 to December 31, 2022. The 2022 election is linked to Lula's third administration. Because the mandate was still ongoing at the time covered by the cabinet data, the analysis observes this mandate from January 1, 2023 through the latest reliable cabinet-period endpoint in the data. In the current source data, this endpoint is March 19, 2026.

[2] Precise definitions are given in Section 3.

[3] These are Axelrod's minimal connected winning coalitions (Axelrod 1970).

the vote, and nevertheless holds 258 seats.

The rest of the article proceeds as follows. Section 2 situates coalition inversions in relation to proportional representation, apportionment, and the literature on election inversions. Section 3 defines the main quantities used in the analysis, including coalition vote share, seat share, quota, seat differential, and the inversion criterion. Section 4 describes the election, cabinet, party-translation, and ideology data. Section 5 presents the party-level seat differentials that enter the coalition analysis. Section 6 presents the observed cabinet-period results for the mandates following the 2014, 2018, and 2022 elections. Section 7 turns to ideological intervals in the election-year party order and identifies minimal connected winning inversions. Section 8 discusses the implications and limitations of the results. The conclusion summarizes the article's contribution and identifies the main extensions.

2 Coalition inversions under Brazilian proportional representation

The literature on proportional representation identifies several ways in which seat allocations can depart from democratic and proportional intuitions. These departures include voting paradoxes in list PR, threshold effects, apportionment effects, spoiler effects, manipulation, unequal representation, and allocation problems internal to party lists or coalitions (Nurmi 1999; Saari 2001; Van Deemen 1993; Balinski and Young 2010; Kaminski 2002; Kaminski 2018; Jaroslaw Flis et al. 2019; Wada and Kamahara 2024). The present analysis isolates one object within that broader family: coalition-level election inversions. The point is not that PR generally fails, but that proportionality evaluated at the party level does not settle how seats behave once parties are recombined into politically relevant sets.

The direct inversion literature begins from the observation that PR can conflict with ordinary majority intuitions. Kurrild-Klitgaard shows, using Danish election data, that proportional representation can generate rank-order discrepancies between seat allocations and majority comparisons over parties (Kurrild-Klitgaard 2008). His later work shifts the problem toward government formation by showing that a party set may form or support a government even when the corresponding coalition does not represent a vote majority (Kurrild-Klitgaard 2013). Along this line, Miller (2014), the main technical predecessor for the present analysis, argues that, in proportional systems, the relevant inversion is usually coalitionwise rather than partywise, since single parties rarely win majorities of both votes and seats. Felsenthal and Miller (2015) then ask what institutional devices might reduce or eliminate such inversions once they are recognized.

This article therefore treats coalition inversions as one PR pathology among others, not as the whole problem of proportional representation. The argument parallels recent work on electoral-system pathologies in mixed systems. Institutions designed to combine proportional and majoritarian virtues can also generate paradoxes, strategic vulnerabilities, overhang-seat problems, compensation failures, and contamination effects (Jarosław Flis, Grofman, et al. 2025; Moon and Kim 2025; Tanács-Mandák and Horváth 2025; Behnke 2025; Jarosław Flis, Behnke, et al. 2025; Vučićević 2025). The present paper makes an analogous move for Brazilian PR. It asks how a system usually defended by party-level proportionality can still generate majority-status distortions once parties are aggregated into cabinets, ideological blocs, or other politically consequential coalitional sets.

siaroff2003spurious distinguishes a manufactured majority, in which the vote-leading party receives less than half of the popular vote but obtains a seat majority, from a spurious majority, in which the seat-majority party is not the popular-vote leader. The distinction also clarifies that a vote-minority seat majority need not itself be anomalous or unintended: thresholds and other electoral rules may deliberately facilitate majority formation at some cost to proportionality. Coalition inversion is a different object. When a coalition C is compared with its complement, a vote-minority seat majority entails a reversal between the two exhaustive party sets, making its structure analogous to the reversal underlying Siaroff's spurious majority. Because Siaroff's categories concern parties, whereas the present analysis concerns post-electoral party sets, I retain the term *coalition inversion* rather than relabeling the Brazilian cases as spurious majorities.

The conceptual point requires care in the Brazilian case because the word "coalition" can refer to different institutional objects. In the abstract PR literature, a coalition is usually a set of parties grouped after seats have been allocated. It is not itself the list to which votes are initially assigned. Brazil, however, has also used collective electoral units. In 2014 and 2018, parties could compete for federal-deputy seats through proportional electoral coalitions (*coligações proporcionais*). These electoral units are distinct from the coalitions analyzed in this article.

The inversion test is applied to post-electoral party sets, not to electoral lists as such. Observed cabinet coalitions are sets of parties that participate in government during a cabinet period. Ideological intervals are analytically constructed sets of parties that are contiguous in an ideological ordering. Both are reconstructed after the election from party-level vote and seat totals. Thus, even when some seats were originally won through *coligações*, the empirical question is whether a later party set, translated into the relevant election-year party labels, crosses the Chamber seat-majority threshold without crossing the national vote-majority threshold.

With this distinction in place, the coalitionwise logic becomes straightforward. A proportional formula may be monotonic with respect to the electoral units to which it allocates seats: if list A receives more votes than list B, the rule will not ordinarily assign fewer seats to A than to B, setting aside thresholds, district effects, and other institutional complications. But this does not imply monotonicity for every post-electoral set of parties. A cabinet coalition or ideological interval may combine parties whose seats were produced through different district contests and, in 2014 and 2018, through different proportional electoral coalitions. A party set can therefore cross the seat-majority threshold even if the same set falls short of a national vote majority.

Spoiler effects provide a second bridge between coalition inversions and the broader literature on PR pathologies. In PR systems, spoilers need not operate only by changing which party wins a plurality. They can operate by removing votes from the seat-allocation stage. When parties fail to cross legal or effective thresholds, their votes do not translate into seats, and the seat shares of the remaining parties are inflated. Kaminski (2018) analysis of Polish parliamentary elections shows that these effects can be large under D'Hondt PR. In the 2015 election, for example, PiS won 37.58 percent of the vote and 51.09 percent of the seats, a seat majority produced in part by votes lost by parties outside the seat distribution. The Brazilian cases analyzed here are different because the object is not a single-party manufactured majority but a post-electoral party set. The mechanism is nevertheless related. Spoilers and wasted votes change the distribution of effective seats across parties, while coalition formation recombines those party-level differentials after the election.

Brazilian proportional representation is not a national list system. Federal deputies are elected in the states and the Federal District, each of which forms a separate multimember district. The Chamber has 513 seats, distributed across these districts according to population, subject to a constitutional floor of eight seats and a ceiling of seventy seats per unit (Câmara dos Deputados 2026). As a result, the national Chamber is not produced by applying a single proportional formula to national party vote totals. It is produced by aggregating twenty-seven district-level proportional allocations.

Within each district, Brazil uses open-list proportional representation. Voters may cast nominal votes for candidates or party-label votes, but these votes are aggregated at the relevant party or joint-list unit for seat allocation. In 2014 and 2018, proportional electoral coalitions pooled party votes within districts. The electoral quotient defines the number of votes corresponding to a district seat; the party quotient determines how many seats the relevant list initially receives; and remaining seats are allocated by averages (Tribunal Superior Eleitoral 2023). Elected candidates are then selected from within the list, subject to individual-vote requirements. The ballot is candidate-centered, but seat allocation is first mediated by collective partisan units.

This electoral structure connects directly to coalitional presidentialism. Since no presidential party normally controls a seat majority alone, presidents must govern by assembling party coalitions. Abranches's formulation of *presidencialismo de coalizão* identifies the combination of presidentialism, multipartism, federalism, proportional representation, and coalition management as a defining feature of Brazilian democracy (Abranches 1988). Figueiredo and Limongi revise the pessimistic interpretation of this arrangement: Brazilian presidents do not simply bargain with atomized deputies, but form governments through parties, distribute ministries, and rely on party leaders in a centralized legislative organization (Figueiredo and Limongi 2000). Pereira and Mueller add a complementary account of executive-legislative exchange through budgetary and distributive resources (Pereira and Mueller 2004).

The implication for this article is that the politically relevant vote-seat conversion is not only the electoral unit's individual conversion, but also the post-electoral party set's aggregate conversion. A governing coalition is assembled from parties whose seats were produced in separate district contests and sometimes through prior electoral groupings. Once those parties are grouped nationally, their seat total may cross the Chamber majority threshold even if their combined national federal-deputy vote does not. The analysis does not ask whether such coalitions are stable, programmatic, or legislatively cohesive. Nor does it infer voter preferences over cabinets from party votes. It asks whether a party set, once translated into the party labels used in the corresponding election returns, receives enough seats to command a Chamber majority while receiving less than half of the national vote in that election.

3 Measurement

Following Miller (2014), let P be the set of parties in a federal-deputy election. For each party $i \in P$, let v_i denote the party's national federal-deputy vote total, let s_i denote the number of Chamber seats won by that party, and let $V = \sum_{i \in P} v_i$ be the total national vote. Since the Chamber has 513 seats, the proportional seat quota of party i is $q_i = 513\frac{v_i}{V}$. The party's seat differential is $d_i = s_i - q_i$.

Positive values mean that the party received more seats than its national vote share would imply under exact proportionality. Negative values mean that the party received fewer. Since quotas sum to 513 and seats sum to 513, party-level seat differentials sum to zero.

For any coalition $C \subseteq P$, define

$$v_C = \sum_{i \in C} v_i, \qquad s_C = \sum_{i \in C} s_i,$$

$$q_C = 513\frac{v_C}{V}, \qquad d_C = s_C - q_C.$$

A coalition has a vote majority when $v_C/V > .5$ and a seat majority when $s_C \geq 257$. A coalition inversion occurs when a party set C has a seat majority but less than half of the vote: $s_C \geq 257$ and $\frac{v_C}{V} < .5$. Equivalently, C is a Chamber majority without being a national federal-deputy vote majority.

4 Data

The empirical analysis combines TSE federal-deputy vote totals, elected-seat totals, cabinet-composition data, and party ideology classifications. Vote and seat data for the 2014, 2018, and 2022 elections come from the Tribunal Superior Eleitoral (TSE), accessed through the `electionsBR` package (Meireles et al. 2016). Cabinet composition is reconstructed from the Brazilian Wikipedia cabinet-member lists for the Dilma Rousseff, Michel Temer, Jair Bolsonaro, and Lula III administrations. These sources identify ministerial appointments by office, incumbent, party label, entry date, and exit date. I use these appointment records to construct cabinet periods, defined by changes in the set of parties holding ministries or ministry-level offices[4].

Cabinet periods are linked to elections by mandate-window overlap. The 2014 election is linked to cabinet periods active from January 1, 2015 through December 31, 2018; the 2018 election to periods active from January 1, 2019 through December 31, 2022; and the 2022 election to the observed Lula III window beginning on January 1, 2023 and continuing through March 19, 2026, the latest reliable cabinet-period endpoint in the source data. Coalition vote and seat totals are computed using the party labels and party-level totals from the election that produced the corresponding Chamber delegation. These inputs are used to construct two analytical objects: observed cabinet-period coalitions and ideological intervals defined over the election-year party set. Each party set is evaluated by its national vote share, seat share, quota, seat differential, and coalition-inversion status.

The national vote measure is the valid federal-deputy vote credited to each party under its election-year label. This includes both nominal votes and party-label votes, including valid nominal votes later treated as party-label votes in the TSE aggregation.[5]

The ideology analysis uses external party classifications to order parties on a left-right scale. The 2014 and 2018 orders use Bolognesi, Ribeiro, and Codato (2023). The 2022 order uses the later 2018–2022 update, operationalized through the 2025 data release and associated article (Bolognesi, Codato, et al. 2025; Bolognesi, Ribeiro, Codato, and Silva 2026). This creates a methodological caveat: the 2014 ordering retrojects later expert classifications onto an earlier party system. Appendix A therefore reports the full party order, classification label, numeric ideology value, source year, and source-party label used for each election.

5 Party-level disproportionality

The coalition analysis begins with party-level seat differentials. Table 1 reports the largest positive and negative differentials by election.

The pattern is not the same across years. In 2014, the largest positive differentials are concentrated in parties that later enter or surround the Temer-period coalitions, especially PMDB, PSD, PTB, PR, and PP. In 2018, several parties associated with the Bolsonaro congressional support base are overrepresented, but the observed cabinet coalitions do not accumulate enough seats to become Chamber majorities. In 2022, the largest positive differentials are associated with UNIÃO and PP, while PL also receives a positive seat differential despite already being the largest party in vote share. This makes the 2022 right-side ideological interval especially consequential.

[4]Further implementation details, including party-label harmonization and translation tables, are documented in the replication repository: https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md.

[5]For 2014, the vote measure is `QT_VOTOS_NOMINAIS + QT_VOTOS_LEGENDA`. For 2018 and 2022, it is `QT_VOTOS_NOMINAIS_VALIDOS + QT_TOTAL_VOTOS_LEG_VALIDOS`.

Table 1: Largest party-level seat differentials by election

Election	Party	Vote share	Seats	Quota	Seat diff.
2014	PMDB	11.09	65	56.87	8.13
2014	PSD	6.13	36	31.45	4.55
2014	PTB	4.02	25	20.63	4.37
2014	PSDB	11.39	54	58.43	-4.43
2014	PSOL	1.79	5	9.20	-4.20
2018	PP	5.50	37	28.24	8.76
2018	PR	5.32	33	27.28	5.72
2018	MDB	5.54	34	28.40	5.60
2018	PSL	11.64	52	59.70	-7.70
2018	NOVO	2.80	8	14.35	-6.35
2022	UNIÃO	8.18	59	41.96	17.04
2022	PP	6.59	47	33.80	13.20
2022	PSD	7.63	42	39.16	2.84
2022	PL	17.96	99	92.13	6.87
2022	PSOL	3.13	12	16.07	-4.07

Notes: Positive values indicate that the party received more seats than its national vote quota. The table only reports selected large positive and negative differentials.

Figure 1 shows how the Chamber translates party vote shares into seat shares across the 2014, 2018, and 2022 elections. Exact proportionality would place every party on the dashed line. Deviations above and below that line are the party-level seat differentials that later enter the coalition analysis. The key point is that these differentials are not exceptional outliers. They are distributed across the party system and appear in each election. Therefore, if a coalition inversion happens, then it is not produced by a single implausible observation. It will be produced when the ordinary arithmetic of representation gives a set of parties a positive aggregate seat differential large enough to cross the legislative majority threshold, despite falling short of a vote majority.

6 Cabinet coalitional inversions

Observed cabinet coalitions are the central empirical object because they are actual party sets. Many possible coalitions can be constructed after any fragmented election. Most are politically meaningless. A cabinet-period coalition is different. It identifies a party set that was institutionally relevant during a specific period of government formation or support.

Table 2 reports the observed cabinet-period inversions[6]. The 2014 mandate is the period in which the composition changes are most substantively important. The initial 2015.1 coalition

[6]Because coalition membership changes over short periods, Appendix B reports every cabinet period, its party composition, and the parties that enter or leave relative to the previous period in the same mandate. These transitions are calculated after translating cabinet labels back into the party labels that existed at the relevant election. For instance, MDB is mapped back to PMDB for the 2014 election, PL is mapped back to PR for the 2014 and 2018 elections, Republicanos is mapped back to PRB when appropriate, and Podemos is mapped back to PTN for the 2014 election. Fusion cases are handled differently from one-to-one renames. In the 2018-linked post-fusion Bolsonaro periods, União Brasil is first used for contemporaneous cabinet-period construction and then expanded into its 2018 election-party antecedents, DEM and PSL, for vote-seat accounting.The classification column uses the following categories. A coalition is *votes+seats* when it has both a vote majority and a seat majority. It is *votes only* when it has a vote majority without a seat majority. It is *seats only* when it has a seat majority without a vote

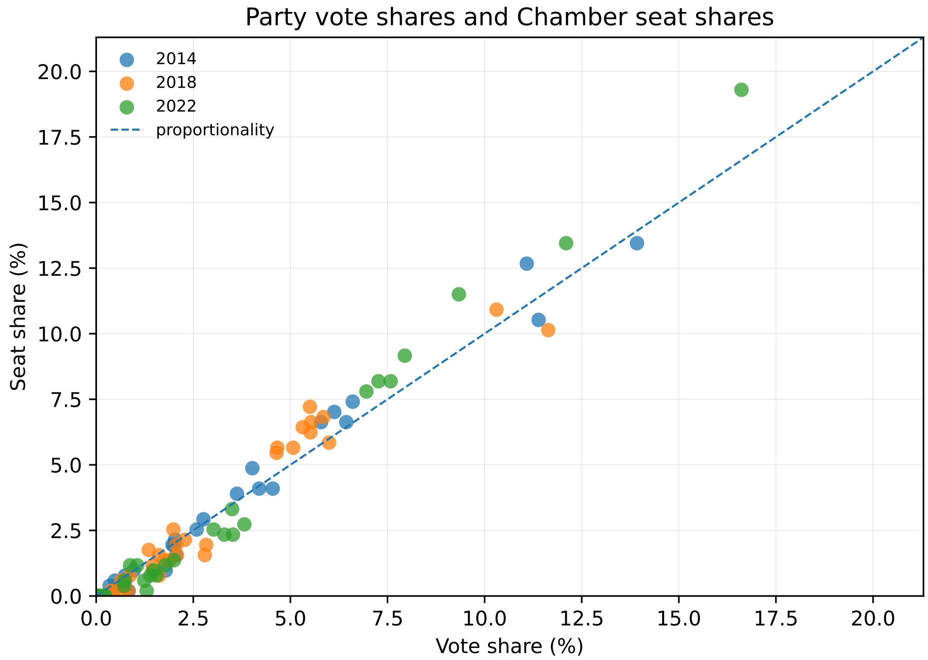

Figure 1: Party vote shares and Chamber seat shares

contains PCdoB, PDT, PMDB, PP, PR, PRB, PSD, PT, and PTB. PRB leaves in 2016.1, PP leaves in the short 2016.2 window, and PSD leaves in 2016.3. The 2016.4 period then marks the post-impeachment reconfiguration: DEM, PP, PPS, PSB, PSD, PSDB, and PV enter, while PCdoB, PDT, PR, and PT leave. PMDB and PTB remain across that rupture. The 2017.1 coalition drops PTB and becomes the most important Temer-period inversion. Later, DEM, PSB, and PV leave in 2018.1, and PTN enters in 2018.2.

The 2018 mandate contains thirteen cabinet periods. PSC enters on December 9, 2020 and remains in the translated election-space coalition through March 29, 2022. After PP enters, the 2021.3 and 2022.1 coalitions each reach exactly 257 seats with 47.25 percent of the vote. PSC leaves on March 30, 2022, ending the inversion. The final 2022 sequence continues to use the União Brasil lineage rule: UNIÃO defines contemporaneous cabinet-period changes and is then expanded back to DEM and PSL for 2018 vote-seat accounting.

The 2022 mandate begins with MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, and UNIÃO. That initial 2023.1 coalition is an inversion. In 2023.2, PP and Republicanos enter, converting the coalition into one with both a vote majority and a seat majority. In 2025.1, UNIÃO leaves the cabinet party set in the available data window; the coalition remains a vote-and-seat majority.

Figure 2 shows why the inversion criterion is more demanding than ordinary overrepresentation. In the 2014-linked mandate, the figure shows the short 2016.2 impeachment-window inversion and the more durable 2017.1 inversion. In the 2018-linked mandate, the PSC-inclusive sequence reaches the seat-majority threshold in 2021.3 and remains there through 2022.1. In the 2022-linked mandate, the initial Lula coalition appears as an inversion, but later cabinet expansion pushes the coalition above both thresholds.

The 2016.2 case should not be dismissed because of its short duration. It is not evidence of a

majority. The last category is the inversion category.

Table 2: Observed cabinet-period coalition inversions

Election	Period	Days	Vote %	Seats	Quota q_C	Diff. d_C	Req. r_C	Parties
2014	2016.2	2	46.54	259	238.73	20.27	18.27	PCdoB, PDT, PMDB, PR, PSD, PT, PTB
2014	2017.1	100	49.91	266	256.05	9.95	0.95	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PV
2018	2021.3	188	47.25	257	242.38	14.62	14.62	DEM, PATRIOTA, PP, PR, PRB, PSC, PSD, PSDB, PSL
2018	2022.1	50	47.25	257	242.38	14.62	14.62	DEM, PATRIOTA, PP, PR, PRB, PSC, PSD, PSDB, PSL
2022	2023.1	255	48.89	263	250.80	12.20	6.20	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO

stable governing arrangement, but it is precisely for that reason an important event-window result. The period falls inside the cabinet realignment sequence surrounding the impeachment process: after PRB and PP leave the cabinet party set, but before the subsequent loss of PSD reduces the coalition below the seat-majority threshold. In that narrow window, the remaining cabinet parties no longer correspond to a national federal-deputy vote majority, with only 46.54 percent of the vote, but they still hold 259 seats. This is the largest observed cabinet inversion in the analysis by seat differential. The result therefore identifies the point in the impeachment sequence at which the representational asymmetry becomes most visible: a cabinet party set associated with less than half of the national federal-deputy vote still retained a Chamber majority.

The 2017.1 case is different. It is not the largest inversion by seat differential, but it is the more durable Temer-period inversion. The coalition composed of DEM, PMDB, PP, PPS, PSB, PSD, PSDB, and PV had 49.91 percent of the national federal-deputy vote and 266 Chamber seats. It therefore crossed the 257-seat threshold while remaining just short of the vote-majority threshold. The 2022 result shows that the phenomenon is not confined to the impeachment period. The initial Lula coalition, from January 1 to September 12, 2023, held 263 seats with 48.89 percent of the national vote. This coalition was therefore also a seat majority without a vote majority. After PP and Republicanos enter the coalition, the status changes: the 2023.2 coalition has 63.79 percent of the vote and 350 seats. It remains overrepresented, but it is no longer an inversion because it has both votes and seats.

The 2018 cases show that the observed-cabinet and ideological-interval domains can diverge. Periods 2021.3 and 2022.1 each have 47.25 percent of the vote, 257 seats, and a seat differential of 14.62 seats, so both are inversions. The 2018 ideological-interval search nevertheless contains zero inversions.

A duration summary helps discipline the interpretation. In the mandate following the 2014 election, 102 of 1461 covered days are inversion days. In the mandate following the 2018 election, 238 of 1461 covered days are inversion days. In the available 2022 window, 255 of 1174 covered days are inversion days. Thus, the observed inversion pattern is not evenly distributed over mandates. It is concentrated in a short transition window and a later Temer period after 2014, the PSC-inclusive phase after 2018, and the initial cabinet formation after 2022.

7 Ideologically contiguous intervals

Are there contiguous ideological blocks of parties that also cross the seat-majority threshold before crossing the vote-majority threshold? To answer this, parties are ordered by ideology and every

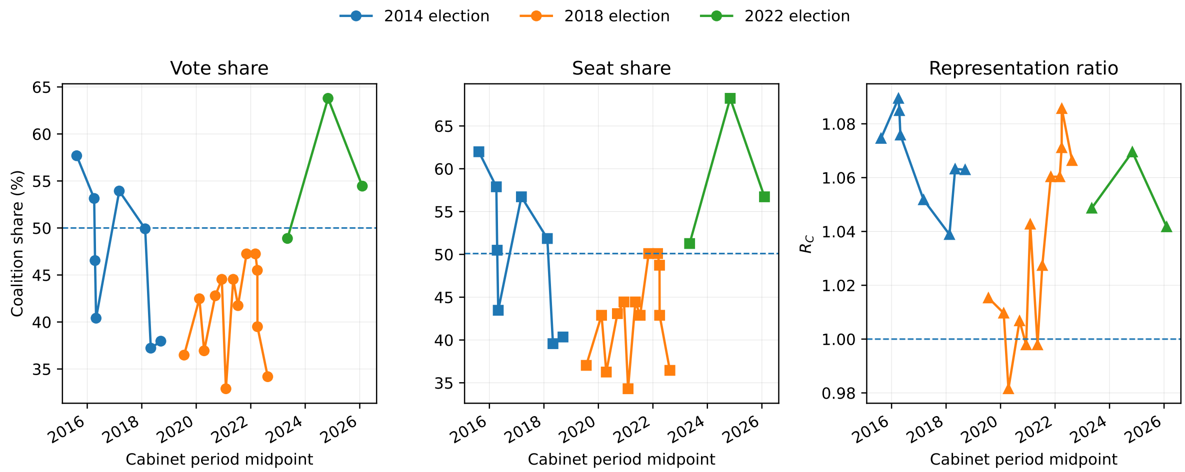

Figure 2: Observed cabinet-period coalition vote shares, seat shares, and representation ratios

Notes: The panels report coalition vote shares, seat shares, and representation ratios. Reference lines mark the vote-majority threshold, the seat-majority threshold of 257/513, and exact proportionality at $R_C = 1$.

contiguous no-gap interval is enumerated. If the ordered parties are $p_1, p_2, \ldots, p_n$, then an interval is any set $[p_i, p_j] = \{p_i, p_{i+1}, \ldots, p_j\}, \quad i \leq j$.

An ideological interval $[p_i, p_j]$ is a minimal connected interval inversion when it is a coalition inversion and removing either endpoint destroys the seat majority. This is minimality under contiguous endpoint deletion. It is precisely what Axelrod (1970) calls minimal connected winning coalitions. This definition is intentionally stricter than a generic search over all party subsets and less restrictive than a search over observed cabinets only. It rules out cherry picked non-contiguous collections, but it does not require that the party set ever formed a cabinet. It shows whether there are seat-vote inversions embedded in the ideological arrangement of parties.

Table 3: Ideological interval summary by election

Election	All intervals	Seat-majority intervals	Inversions	Minimal inversions
2014	528	180	8	4
2018	630	172	0	0
2022	528	99	6	2

Table 3 reports the aggregate pattern. The first count is mechanical: with n parties in the compact ideology order, there are $n(n+1)/2$ contiguous intervals. The larger 2018 count therefore reflects the larger mapped party set in that election, not a substantive increase in observed coalitions. The substantive comparison begins with the seat-majority and inversion columns. In 2018, 172 contiguous intervals reach a seat majority, but none are inversions. Thus, the absence of 2018 interval inversions is not due to a lack of large ideological blocs. It reflects the vote-seat relationship itself: the seat-majority intervals in that year also cross the vote-majority threshold.

The minimal intervals in Table 4 reveal two different forms of the phenomenon. The 2014 intervals are close to the 50 percent vote threshold and often just at the 257-seat threshold. These are genuine inversions, but they are close-run cases in the sense expected by the theory of PR

Table 4: Minimal connected interval inversions

Election	Start	End	Parties	Vote share	Seats	Seat diff.	Interval
2014	PSB	PTN	12	48.92	260	9.03	PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN
2014	PTB	PR	14	47.59	257	12.86	PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR
2014	PT DO B	PSDC	16	48.99	257	5.70	PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC
2014	SOLIDARIEDADE	PSL	16	48.98	257	5.72	SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL
2022	MDB	UNIÃO	15	49.98	265	8.59	MDB, PMN, PSDB, PSD, PMB, PODE, PROS, PRTB, AGIR, PTB, PP, DC, REPUBLICANOS, PSC, UNIÃO
2022	PP	PL	8	45.35	258	25.36	PP, DC, REPUBLICANOS, PSC, UNIÃO, PATRIOTA, NOVO, PL

Notes: The Parties column reports interval size. Vote share is percentage of national federal-deputy votes.

inversions (Miller 2014). The 2022 PP–PL interval is different. It is compact, right-side, and substantially below the vote-majority threshold. With only 45.35 percent of the vote, it holds 258 seats. This is the strongest structural result in the analysis of minimal winning coalitions done here.

Figure 3 gives the interval analysis its substantive content. Each heatmap shows all contiguous ideological intervals in the Chamber delegation. A cell is defined by the interval's left endpoint and right endpoint in the party order. Blue cells identify intervals that reach the Chamber majority threshold without satisfying the inversion criterion. Orange and red cells identify the stronger event: intervals that reach a seat majority while remaining below a national federal-deputy vote majority.

The distinction between orange and red cells is important. An inversion is any contiguous interval with at least 257 seats and no vote majority. A minimal inversion is an inversion that ceases to be seat-winning if either endpoint is removed. Minimality identifies the critical connected bloc at which a vote-minority interval first becomes a Chamber majority. Larger orange regions may be genuine inversions, but many of them are expansions of a smaller red interval. The red cells therefore mark the frontier of coalitional seat efficiency inside the ideological order. Appendix C reports the minimal connected winning intervals, and the red cells are the subset of those intervals that also satisfy the inversion criterion.

The main pattern is that the ideological location and compactness of inversions change. In 2014, the inversion intervals are broad and cross the center. One minimal connected coalitional inversion begins with PSB, a center-left party in the ideology classification, and runs through PPS and PV before accumulating a sequence of center-right and right parties. The other minimal connected coalitional inversions begin farther to the right, but they still belong to an extended center-to-right region rather than to a compact bloc at the right edge of the party system. The 2014 pattern

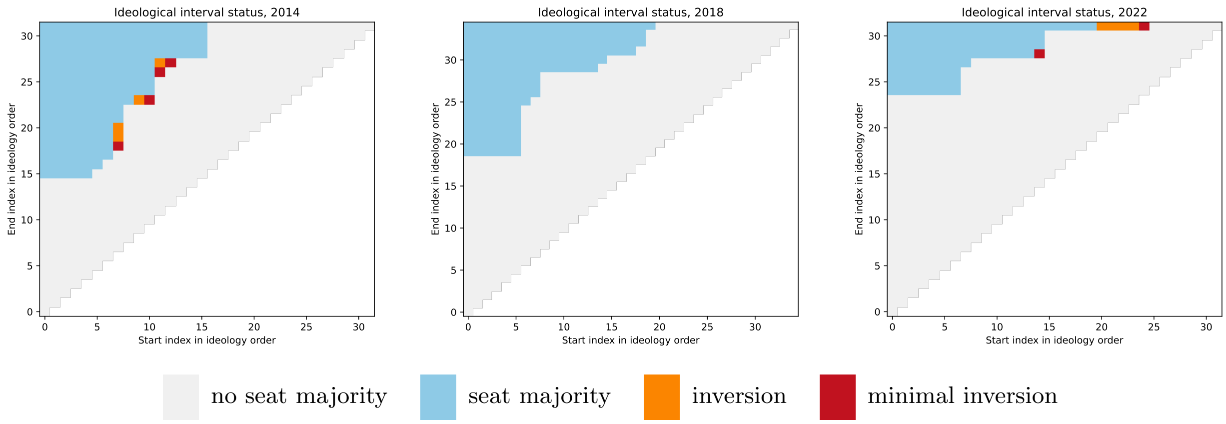

Figure 3: Ideological interval status by election

Notes: Each matrix enumerates contiguous ideological intervals in the election-year party order by start and end index.

therefore suggests a diffuse form of coalitional seat efficiency: inversions are produced by broad intervals that gather enough parties across the middle and right of the ideological order.

The 2022 panel is different. The seat-majority intervals are more compressed, and the inversion cells are displaced sharply to the right. The broad MDB–UNIÃO interval begins in the center-right and ends in the far right. More importantly, the PP–PL interval is entirely located on the right side of the ideological order. It begins with PP, classified as right, and ends with PL, classified as far right, passing through DC, REPUBLICANOS, PSC, UNIÃO, PATRIOTA, and NOVO. This interval is not only compact. It is ideologically one-sided. A bloc located at the right and far-right edge of the Chamber can reach 258 seats with only 45.35 percent of the national federal-deputy vote. The result is stronger than the mere presence of orange cells in the heatmap: the minimal source of the inversion is already a narrow right-side bloc.

The figure also clarifies the weakness of the left in the Chamber. Left and center-left parties do not form an autonomous seat-majority region in any of the three panels. Intervals that begin on the left become seat majorities only after they expand far enough into the center, center-right, or right. In this sense, the left appears as the endpoint of broad coalitions, not as the core of a compact seat-majority bloc. By contrast, in 2022 the right does form such a core: the PP–PL interval shows that a relatively narrow right/far-right bloc can cross the majority threshold without a corresponding vote majority.

The Appendix D table reinforces this reading and adds an important political qualification. PT held the presidency during part of the mandate associated with the 2014 election, through Dilma Rousseff's second administration, and again in the mandate associated with the 2022 election, through Lula's third administration. For that reason, PT is not just another party in the interval order. It is the presidential party in two of the three election-linked periods covered by the analysis. Yet PT appears in only four of the twenty minimal connected winning intervals: two in 2014, one in 2018, and one in 2022. All four are vote-and-seat majorities, not inversions. They are also broad intervals. The two PT-containing intervals in 2014 contain 12 parties, the 2018 interval contains 15 parties, and the 2022 interval contains 19 parties. This is a relevant contrast: minimal connected majorities that include the presidential party require wide ideological spans and vote majorities, while the vote-minority seat-majority intervals are located farther to the right. The compact 2022

PP–PL interval is therefore not just another minimal winning interval. It identifies a right-side bloc whose seat conversion is strong enough to produce a Chamber majority without a national vote majority.

The 2018 panel isolates a useful null result within the ideological-interval domain. It contains many seat-majority intervals, but none of them are inversions. Contiguous ideological blocks could become legislative majorities, but only after also crossing the vote-majority threshold. This indicates that the absence of 2018 inversions is not due to the absence of large contiguous ideological intervals. It is due to the fact that the vote-seat conversion did not create a contiguous ideological bloc that was both seat-majoritarian and vote-minoritarian.

Taken together, the three panels show a change in the ideological structure of coalitional overrepresentation. In 2014, inversion possibilities are distributed across broad center-to-right intervals. In 2018, seat-majority intervals remain above the vote-majority threshold. In 2022, the inversion intervals move to the right edge of the party system and become more compact. The Chamber's right flank becomes sufficiently seat-efficient that a compact right/far-right interval can command a legislative majority without a national vote majority, while minimal connected majorities that include the left require broader vote-majority coalitions.

The connection between the cabinet results and the interval results is developed in Appendix D. The key point is that observed cabinets are not usually contiguous ideological blocs. They are sparse selections from wider ideological spans. The Dilma coalitions are left-anchored but punctured: they include PT, PCdoB, and PDT while extending into PMDB, PSD, PR, PP, and PTB. The post-impeachment Temer coalitions remain punctured but shift rightward, replacing the left anchor with a center-to-right configuration. The 2023 Lula coalition returns to a left-anchored but very broad structure, while the compact connected inversion in 2022 is not the Lula cabinet but the right-side PP–PL interval. Thus, the cabinet analysis identifies realized governing coalitions, while the interval analysis shows where seat-efficient connected blocs are located in the ideological order.

8 Discussion

The analysis identifies a narrow but politically significant form of representational inversion. A coalition inversion does not mean that proportional representation has failed as a party-level allocation rule. It means that a party set has been converted into a legislative majority without corresponding to a national federal-deputy vote majority. In a fragmented legislature, this is not a secondary issue. Cabinets, support bases, and ideological blocs are constructed from party sets, not from isolated parties.

This distinction gives the Brazilian case its institutional importance. The standard evaluation of proportional representation asks whether parties receive seats in proportion to votes. That question remains relevant, but it is incomplete under coalitional presidentialism. Once presidents govern by assembling parties, the representational object becomes composite. A cabinet coalition can inherit the accumulated seat surplus of its member parties and thereby cross the Chamber-majority threshold before it crosses the vote-majority threshold.

The 2014-linked mandate illustrates this possibility most sharply. The short 2016.2 period falls inside the impeachment realignment and is not evidence of a stable cabinet configuration. Its importance lies elsewhere. During that realignment, the remaining cabinet party set no longer corresponded to a national federal-deputy vote majority, but it still held a Chamber majority. The 2017.1 Temer coalition is the more durable case: 266 seats with 49.91 percent of the national federal-deputy vote. This case gives a precise institutional form to the representational problem:

the governing arrangement was supported by a congressional majority whose seat status exceeded its national vote basis.

The 2018-linked mandate distinguishes the two coalition domains. PSC-inclusive Bolsonaro cabinets in 2021.3 and 2022.1 each reach exactly 257 seats with 47.25 percent of the vote, while none of the contiguous ideological intervals in the 2018 party order is an inversion. The result is a within-election contrast between realized, sparse cabinet coalitions and connected ideological blocs.

The 2022-linked mandate complicates the political interpretation in the opposite direction. Lula's initial cabinet is also an observed inversion. It begins as a seat majority without a national federal-deputy vote majority, then becomes a vote-and-seat majority after PP and Republicanos enter. But the ideological interval analysis shows a different structure beneath the cabinet sequence. The most striking connected inversion is not the Lula coalition. It is the PP–PL interval, a compact right and far-right bloc that controls 258 seats with only 45.35 percent of the national vote.

This contrast is central. The observed cabinet inversions show that vote-minority seat-majorities can appear in actual government formation, while the ideological interval inversions show where seat-efficient connected blocs are located in the party system. In 2014, the inversion intervals are broad and center-to-right. In 2022, they become more compact and move to the right edge of the Chamber. The left appears as an endpoint of broad coalitions. The right, by contrast, can form a narrower connected bloc with majority seat status. Thus, Brazil's open-list PR institutional matrix can allocate seats in such a way that coalitional recombination creates Chamber majorities not backed by national vote majorities.

This is why the Brazilian case belongs in the broader discussion of the dark side of proportional representation. The analysis does not show that PR is generally inferior to majoritarian systems, nor that Brazilian representation is simply disproportional. It shows a more specific institutional pathology. A system may be evaluated as proportional at the party level while still producing majority-status distortions at the level where legislative authority is exercised. In that sense, coalition inversions are parallel to spoilers, threshold effects, apportionment effects, mixed-system paradoxes, and manipulation problems: each identifies a point at which the electoral rule translates votes into politically consequential seats in a way that is not visible from simple party vote shares alone (Van Deemen 1993; Kurrild-Klitgaard 2008; Kaminski 2018; Jarosław Flis, Grofman, et al. 2025).

The current pipeline decomposes all five observed inversions as an accounting identity, $d_C = A_C + B_C$, where A_C is the within-district allocation component and B_C is the between-district weighting component. The latter combines district seat weights, turnout and valid-vote differences, and coalition vote geography; it is not pure malapportionment and the identity is not causal. The authoritative review manuscript reports the full decomposition and party-contribution results.

The ideological analysis should also be treated as a structured diagnostic rather than a final map of Brazilian party ideology. The classifications make it possible to enumerate connected intervals, but the 2014 and 2018 orders depend on later classification sources and party-label translation. Alternative placements may shift endpoint labels, interval widths, or marginal cases.

The last limitation concerns interpretation. The vote share of a cabinet or interval is not a direct vote for that cabinet or interval. It is the sum of votes cast for parties later grouped into that set. The criterion therefore measures coalitional representation, not voter endorsement. It also does not model legislative discipline, roll-call support, or portfolio bargaining. These are behavioral questions. The present object is prior to them: whether the electoral translation gives a party set majority status in the Chamber.

9 Conclusion

The analysis began from a simple problem inside proportional representation. Proportionality is usually evaluated at the level of parties: how many votes a party receives, how many seats it obtains, and how far that seat total is from its proportional quota. But governments are not normally single-party objects in Brazil. They are assembled from party sets. The question, then, is whether a system that is evaluated party by party also preserves majority status once parties are recombined into governing coalitions or ideological blocs.

The answer is not uniform across elections or coalition domains. After the 2014 election, cabinet inversions appear in the impeachment realignment and again in the Temer period. After the 2018 election, PSC-inclusive cabinets generate two inversions at exactly 257 seats, while the ideological-interval result remains null. After the 2022 election, Lula's initial cabinet begins as a seat majority without a vote majority, but later becomes a vote-and-seat majority after cabinet expansion. Connected ideological vote-minority seat-majorities exist in 2014 and 2022, and the strongest 2022 case is the compact PP–PL right-side interval.

These findings show, in the Brazilian context, that seat differentials are not politically inert. Once parties are grouped into cabinets or connected ideological intervals, those differentials can accumulate enough to change majority status. In some configurations, a party set that falls short of half of the national federal-deputy vote nevertheless controls a majority of Chamber seats.

The district accounting decomposition partitions every observed inversion into within-district allocation and between-district weighting components and verifies that party contributions sum to the coalition differential. Explaining the causal roles of apportionment, district magnitude, turnout, party geography, threshold rules and electoral coalitions remains a separate counterfactual research agenda.

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A Party ideology classifications used for interval construction

The interval analysis uses the following source-year rule: 2014 uses the 2023 classification, 2018 uses the 2023 classification, and 2022 uses the 2025 classification. The 2023 classification corresponds to the expert-survey article by Bolognesi, Ribeiro, and Codato (Bolognesi, Ribeiro, and Codato 2023). The 2025 classification file corresponds to the later 2018–2022 classification database and associated Opinião Pública article (Bolognesi, Codato, et al. 2025; Bolognesi, Ribeiro, Codato, and Silva 2026).

For the 2014 ideology join, four label overrides are used: AVANTE → PT DO B, DC → PSDC, MDB → PMDB, and PODE → PTN. do not belong to the 2014 party set used in the vote-seat data: NOVO, PMB, and REDE. No label overrides or source-year drops are applied to the 2018 or 2022 ideology joins in the current runner output.

Table 5 reports the full party order used to build the ideological intervals. The order is generated from the source-year classification file listed in the Source column, after the party-label translations described in the main text. The Source-party column preserves the party label in the source classification before translation into the election-year party label. The classification labels in the table have been translated into English from the Portuguese labels given by the experts.

Table 5: Party ideology order by election

Election	Pos.	Party	Classification	Score	Source	Source party
2014	1	PSTU	far left	0.51	2023	PSTU
2014	2	PCO	far left	0.61	2023	PCO
2014	3	PCB	far left	0.91	2023	PCB
2014	4	PSOL	far left	1.28	2023	PSOL
2014	5	PCdoB	left	1.92	2023	PCdoB
2014	6	PT	left	2.97	2023	PT
2014	7	PDT	center-left	3.92	2023	PDT
2014	8	PSB	center-left	4.05	2023	PSB
2014	10	PPS	center	4.92	2023	PPS
2014	11	PV	center	5.29	2023	PV
2014	12	PTB	center-right	6.10	2023	PTB

Election	Pos.	Party	Classification	Score	Source	Source party
2014	13	PT DO B	center-right	6.32	2023	Avante
2014	14	SOLIDARIEDADE	center-right	6.50	2023	SDD
2014	15	PMN	center-right	6.88	2023	PMN
2014	17	PHS	center-right	6.96	2023	PHS
2014	18	PMDB	right	7.01	2023	MDB
2014	19	PSD	right	7.09	2023	PSD
2014	20	PSDB	right	7.11	2023	PSDB
2014	21	PTN	right	7.24	2023	Podemos
2014	22	PPL	right	7.27	2023	PPL
2014	23	PRTB	right	7.45	2023	PRTB
2014	24	PROS	right	7.47	2023	Pros
2014	25	PRP	right	7.59	2023	PRP
2014	26	PR	right	7.78	2023	PR
2014	27	PRB	right	7.78	2023	PRB
2014	28	PTC	right	7.86	2023	PTC
2014	29	PSDC	right	8.11	2023	DC
2014	30	PSL	right	8.11	2023	PSL
2014	32	PP	right	8.20	2023	Progressistas
2014	33	PSC	right	8.33	2023	PSC
2014	34	PEN	far right	8.55	2023	Patriota
2014	35	DEM	far right	8.57	2023	DEM
2018	1	PSTU	far left	0.51	2023	PSTU
2018	2	PCO	far left	0.61	2023	PCO
2018	3	PCB	far left	0.91	2023	PCB
2018	4	PSOL	far left	1.28	2023	PSOL
2018	5	PCdoB	left	1.92	2023	PCdoB
2018	6	PT	left	2.97	2023	PT
2018	7	PDT	center-left	3.92	2023	PDT
2018	8	PSB	center-left	4.05	2023	PSB
2018	9	REDE	center	4.77	2023	Rede
2018	10	PPS	center	4.92	2023	PPS
2018	11	PV	center	5.29	2023	PV
2018	12	PTB	center-right	6.10	2023	PTB
2018	13	AVANTE	center-right	6.32	2023	Avante
2018	14	SOLIDARIEDADE	center-right	6.50	2023	SDD
2018	15	PMN	center-right	6.88	2023	PMN
2018	16	PMB	center-right	6.90	2023	PMB
2018	17	PHS	center-right	6.96	2023	PHS
2018	18	MDB	right	7.01	2023	MDB
2018	19	PSD	right	7.09	2023	PSD
2018	20	PSDB	right	7.11	2023	PSDB
2018	21	PODE	right	7.24	2023	Podemos
2018	22	PPL	right	7.27	2023	PPL
2018	23	PRTB	right	7.45	2023	PRTB
2018	24	PROS	right	7.47	2023	Pros

Election	Pos.	Party	Classification	Score	Source	Source party
2018	25	PRP	right	7.59	2023	PRP
2018	26	PR	right	7.78	2023	PR
2018	27	PRB	right	7.78	2023	PRB
2018	28	PTC	right	7.86	2023	PTC
2018	29	DC	right	8.11	2023	DC
2018	30	PSL	right	8.11	2023	PSL
2018	31	NOVO	right	8.13	2023	Novo
2018	32	PP	right	8.20	2023	Progressistas
2018	33	PSC	right	8.33	2023	PSC
2018	34	PATRIOTA	far right	8.55	2023	Patriota
2018	35	DEM	far right	8.57	2023	DEM
2022	1	PSTU	far left	0.53	2025	PSTU
2022	2	PCO	far left	0.57	2025	PCO
2022	3	PCB	far left	0.71	2025	PCB
2022	4	PSOL	far left	1.45	2025	PSOL
2022	5	UP	left	1.68	2025	UP
2022	6	PCdoB	left	1.83	2025	PCdoB
2022	7	PT	left	2.76	2025	PT
2022	8	PSB	center-left	3.70	2025	PSB
2022	9	REDE	center-left	3.80	2025	REDE
2022	10	PDT	center-left	3.98	2025	PDT
2022	11	PV	center-left	4.25	2025	PV
2022	12	SOLIDARIEDADE	center-right	6.19	2025	SDD
2022	13	CIDADANIA	center-right	6.36	2025	CDD
2022	14	AVANTE	center-right	6.67	2025	AVANTE
2022	15	MDB	center-right	6.70	2025	MDB
2022	16	PMN	center-right	6.95	2025	PMN
2022	17	PSDB	center-right	6.97	2025	PSDB
2022	18	PSD	right	7.15	2025	PSD
2022	19	PMB	right	7.51	2025	PMB
2022	20	PODE	right	7.67	2025	PODE
2022	21	PROS	right	7.68	2025	PROS
2022	22	PRTB	right	7.72	2025	PRTB
2022	23	AGIR	right	7.78	2025	AGIR
2022	24	PTB	right	7.96	2025	PTB
2022	25	PP	right	8.40	2025	PROGRE
2022	26	DC	right	8.46	2025	DC
2022	27	REPUBLICANOS	far right	8.58	2025	REP
2022	28	PSC	far right	8.67	2025	PSC
2022	29	UNIÃO	far right	8.75	2025	UNIAO
2022	30	PATRIOTA	far right	8.86	2025	PATRI
2022	31	NOVO	far right	8.93	2025	NOVO
2022	32	PL	far right	9.07	2025	PL

B Cabinet periods and composition

Table 6: Cabinet-period composition and transitions after cabinet-label harmonization and election-year label translation

Election	Period	Dates	Days	Vote %	Seats	Seat diff.	Status	Composition	Entered	Left
2014	2015.1	2015-01-01–2016-03-23	448	57.69	318	22.07	votes+seats	PCdoB, PDT, PMDB, PP, PR, PRB, PSD, PT, PTB		
2014	2016.1	2016-03-24–2016-04-13	21	53.14	297	24.39	votes+seats	PCdoB, PDT, PMDB, PP, PR, PSD, PT, PTB		PRB
2014	2016.2	2016-04-14–2016-04-15	2	46.54	259	20.27	seats only	PCdoB, PDT, PMDB, PR, PSD, PT, PTB		PP
2014	2016.3	2016-04-16–2016-05-11	26	40.41	223	15.72	neither	PCdoB, PDT, PMDB, PR, PT, PTB		PSD
2014	2016.4	2016-05-12–2017-12-27	595	53.93	291	14.33	votes+seats	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PTB, PV	DEM, PP, PPS, PSB, PSD, PSDB, PV	PCdoB, PDT, PR, PT
2014	2017.1	2017-12-28–2018-04-06	100	49.91	266	9.95	seats only	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PV		PTB
2014	2018.1	2018-04-07–2018-05-27	51	37.22	203	12.07	neither	PMDB, PP, PPS, PSD, PSDB		DEM, PSB, PV
2014	2018.2	2018-05-28–2018-12-31	218	37.96	207	12.26	neither	PMDB, PP, PPS, PSD, PSDB, PTN	PTN	
2018	2019.1	2019-01-01–2020-02-10	406	36.48	190	2.86	neither	DEM, MDB, NOVO, PATRIOTA, PR, PRB, PSL		
2018	2020.1	2020-02-11–2020-02-14	4	42.47	220	2.11	neither	DEM, MDB, NOVO, PATRIOTA, PR, PRB, PSDB, PSL	PSDB	
2018	2020.2	2020-02-15–2020-06-16	123	36.94	186	-3.50	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSDB, PSL		MDB
2018	2020.3	2020-06-17–2020-12-08	175	42.79	221	1.49	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSD, PSDB, PSL	PSD	
2018	2020.4	2020-12-09–2020-12-09	1	44.54	228	-0.49	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSC, PSD, PSDB, PSL	PSC	
2018	2020.5	2020-12-10–2021-03-28	109	32.90	176	7.21	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSC, PSD, PSDB		PSL
2018	2021.1	2021-03-29–2021-06-23	87	44.54	228	-0.49	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSC, PSD, PSDB, PSL	PSL	
2018	2021.2	2021-06-24–2021-08-03	41	41.74	220	5.86	neither	DEM, PATRIOTA, PR, PRB, PSC, PSD, PSDB, PSL		NOVO
2018	2021.3	2021-08-04–2022-02-07	188	47.25	257	14.62	seats only	DEM, PATRIOTA, PP, PR, PRB, PSC, PSD, PSDB, PSL	PP	
2018	2022.1	2022-02-08–2022-03-29	50	47.25	257	14.62	seats only	DEM, PATRIOTA, PP, PR, PRB, PSC, PSD, PSDB, PSL		
2018	2022.2	2022-03-30–2022-03-31	2	45.50	250	16.60	neither	DEM, PATRIOTA, PP, PR, PRB, PSD, PSDB, PSL		PSC
2018	2022.3	2022-04-01–2022-04-01	1	39.50	220	17.36	neither	DEM, PATRIOTA, PP, PR, PRB, PSD, PSL		PSDB
2018	2022.4	2022-04-02–2022-12-31	274	34.18	187	11.64	neither	DEM, PATRIOTA, PP, PRB, PSD, PSL		PR
2022	2023.1	2023-01-01–2023-09-12	255	48.89	263	12.20	seats only	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO		
2022	2023.2	2023-09-13–2025-12-23	833	63.79	350	22.76	votes+seats	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS, UNIÃO	PP, REPUBLICANOS	
2022	2025.1	2025-12-24–2026-03-19	86	54.45	291	11.66	votes+seats	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS		UNIÃO

Notes: Composition, Entered, and Left are reported using the election-year party labels used for vote-seat accounting. Period windows are constructed from the party labels observed in the cabinet data. When a party fusion occurs, the successor party is used to determine whether the cabinet party set changes; it is then expanded into antecedent election parties if the relevant election occurred before the fusion.

C Minimal connected winning ideological intervals

Table 7 reports all minimal connected winning intervals in the ideology order. These are contiguous intervals that reach the Chamber majority threshold and cease to do so when either endpoint is removed. The table is broader than Table 4, which reports only minimal connected interval inversions. The distinction matters because a minimal connected winning interval may have both a vote majority and a seat majority, while an interval inversion has a seat majority without a vote majority.

Table 7: Minimal connected winning ideological intervals

Election	Start	End	Parties	Vote %	Seats	Seat diff.	Status	Contains PT	Interval
2014	PCdoB	PMDB	12	50.17	265	7.63	votes+seats	Yes	PCdoB, PT, PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB
2014	PT	PSD	12	54.34	291	12.26	votes+seats	Yes	PT, PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD
2014	PDT	PSDB	12	51.80	276	10.25	votes+seats	No	PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB
2014	PSB	PTN	12	48.92	260	9.03	seats only	No	PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN
2014	PTB	PR	14	47.59	257	12.86	seats only	No	PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR
2014	PT DO B	PSDC	16	48.99	257	5.70	seats only	No	PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC
2014	SOLIDARIEDADE	PSL	16	48.98	257	5.72	seats only	No	SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL
2014	PMDB	PP	14	51.37	272	8.45	votes+seats	No	PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL, PP
2018	PT	PSDB	15	50.18	267	9.56	votes+seats	Yes	PT, PDT, PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PMB, PHS, MDB, PSD, PSDB
2018	PDT	PR	20	51.52	268	3.70	votes+seats	No	PDT, PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PMB, PHS, MDB, PSD, PSDB, PODE, PPL, PRTB, PROS, PRP, PR
2018	PSB	PRB	20	51.95	269	2.50	votes+seats	No	PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PMB, PHS, MDB, PSD, PSDB, PODE, PPL, PRTB, PROS, PRP, PR, PRB
2018	SOLIDARIEDADE	PSL	17	51.05	262	0.09	votes+seats	No	SOLIDARIEDADE, PMN, PMB, PHS, MDB, PSD, PSDB, PODE, PPL, PRTB, PROS, PRP, PR, PRB, PTC, DC, PSL
2018	PMN	NOVO	17	51.86	257	-9.06	votes+seats	No	PMN, PMB, PHS, MDB, PSD, PSDB, PODE, PPL, PRTB, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO
2018	MDB	PP	15	55.04	285	2.67	votes+seats	No	MDB, PSD, PSDB, PODE, PPL, PRTB, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP
2018	PSD	PSC	15	51.25	258	-4.91	votes+seats	No	PSD, PSDB, PODE, PPL, PRTB, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP, PSC
2018	PSDB	DEM	16	51.52	257	-7.29	votes+seats	No	PSDB, PODE, PPL, PRTB, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP, PSC, PATRIOTA, DEM
2022	PT	PP	19	57.86	284	-12.81	votes+seats	Yes	PT, PSB, REDE, PDT, PV, SOLIDARIEDADE, CIDADANIA, AVANTE, MDB, PMN, PSDB, PSD, PMB, PODE, PROS, PRTB, AGIR, PTB, PP
2022	PSB	PSC	21	54.58	261	-19.01	votes+seats	No	PSB, REDE, PDT, PV, SOLIDARIEDADE, CIDADANIA, AVANTE, MDB, PMN, PSDB, PSD, PMB, PODE, PROS, PRTB, AGIR, PTB, PP, DC, REPUBLICANOS, PSC
2022	MDB	UNIÃO	15	49.98	265	8.59	seats only	No	MDB, PMN, PSDB, PSD, PMB, PODE, PROS, PRTB, AGIR, PTB, PP, DC, REPUBLICANOS, PSC, UNIÃO
2022	PP	PL	8	45.35	258	25.36	seats only	No	PP, DC, REPUBLICANOS, PSC, UNIÃO, PATRIOTA, NOVO, PL

D Cabinet coalitions and ideological intervals

Table 8 compares observed cabinet coalitions with the ideological intervals that contain or approximate them. The bridge table keeps two operations separate: cabinet periods are defined from the party labels observed in the cabinet data, while ideological closures and vote-seat calculations use the translated election-year party set. The interval analysis therefore remains tied to the Chamber delegation produced by each election. The connected closure of a cabinet is the smallest no-gap ideological interval containing all cabinet parties. Gaps are parties inside that closure that are not members of the cabinet coalition. The final columns report the closest minimal connected winning interval and, where one exists, the closest minimal connected inversion. This table links the observed coalition analysis to the interval analysis by showing whether realized cabinet coalitions are compact ideological blocs or sparse selections from a wider ideological span.

Table 8: Cabinet coalitions and ideological intervals

Election	Period	Cabinet	Ideology summary	Span	Gaps	Closure summary	Minimal winning	Minimal inversion
2014	2015.1	votes+seats; 57.69%; 318 seats	left: 2; center-left: 1; center-right: 1; right: 5	PCdoB–PP	16	left: 2; center-left: 2; center: 2; center-right: 5; right: 14	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.31	PTB–PR; 47.59%; 257 seats; overlap 4; J=0.21
2014	2016.1	votes+seats; 53.14%; 297 seats	left: 2; center-left: 1; center-right: 1; right: 4	PCdoB–PP	17	left: 2; center-left: 2; center: 2; center-right: 5; right: 14	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.33	PTB–PR; 47.59%; 257 seats; overlap 4; J=0.22
2014	2016.2	seats only; 46.54%; 259 seats	left: 2; center-left: 1; center-right: 1; right: 3	PCdoB–PR	13	left: 2; center-left: 2; center: 2; center-right: 5; right: 9	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.36	PTB–PR; 47.59%; 257 seats; overlap 4; J=0.24
2014	2016.3	neither; 40.41%; 223 seats	left: 2; center-left: 1; center-right: 1; right: 2	PCdoB–PR	14	left: 2; center-left: 2; center: 2; center-right: 5; right: 9	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.38	PTB–PR; 47.59%; 257 seats; overlap 3; J=0.18
2014	2016.4	votes+seats; 53.93%; 291 seats	center-left: 1; center: 2; center-right: 1; right: 4; far right: 1	PSB–DEM	16	center-left: 1; center: 2; center-right: 5; right: 15; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 7; J=0.50	PSB–PTN; 48.92%; 260 seats; overlap 7; J=0.50
2014	2017.1	seats only; 49.91%; 266 seats	center-left: 1; center: 2; right: 4; far right: 1	PSB–DEM	17	center-left: 1; center: 2; center-right: 5; right: 15; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 6; J=0.43	PSB–PTN; 48.92%; 260 seats; overlap 6; J=0.43
2014	2018.1	neither; 37.22%; 203 seats	center: 1; right: 4	PPS–PP	16	center: 2; center-right: 5; right: 14	PSB–PTN; 48.92%; 260 seats; overlap 4; J=0.31	PSB–PTN; 48.92%; 260 seats; overlap 4; J=0.31
2014	2018.2	neither; 37.96%; 207 seats	center: 1; right: 5	PPS–PP	15	center: 2; center-right: 5; right: 14	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.38	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.38
2018	2019.1	neither; 36.48%; 190 seats	right: 5; far right: 2	MDB–DEM	11	right: 16; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 6; J=0.35	none
2018	2020.1	neither; 42.47%; 220 seats	right: 6; far right: 2	MDB–DEM	10	right: 16; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 7; J=0.41	none
2018	2020.2	neither; 36.94%; 186 seats	right: 5; far right: 2	PSDB–DEM	9	right: 14; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 7; J=0.44	none
2018	2020.3	neither; 42.79%; 221 seats	right: 6; far right: 2	PSD–DEM	9	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 7; J=0.41	none
2018	2020.4	neither; 44.54%; 228 seats	right: 7; far right: 2	PSD–DEM	8	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 8; J=0.47	none
2018	2020.5	neither; 32.90%; 176 seats	right: 6; far right: 2	PSD–DEM	9	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 7; J=0.41	none
2018	2021.1	neither; 44.54%; 228 seats	right: 7; far right: 2	PSD–DEM	8	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 8; J=0.47	none
2018	2021.2	neither; 41.74%; 220 seats	right: 6; far right: 2	PSD–DEM	9	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 7; J=0.41	none
2018	2021.3	seats only; 47.25%; 257 seats	right: 7; far right: 2	PSD–DEM	8	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 8; J=0.47	none
2018	2022.1	seats only; 47.25%; 257 seats	right: 7; far right: 2	PSD–DEM	8	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 8; J=0.47	none
2018	2022.2	neither; 45.50%; 250 seats	right: 6; far right: 2	PSD–DEM	9	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 7; J=0.41	none
2018	2022.3	neither; 39.50%; 220 seats	right: 5; far right: 2	PSD–DEM	10	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 6; J=0.35	none
2018	2022.4	neither; 34.18%; 187 seats	right: 4; far right: 2	PSD–DEM	11	right: 15; far right: 2	PSDB–DEM; 51.52%; 257 seats; overlap 5; J=0.29	none
2022	2023.1	seats only; 48.89%; 263 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 1; far right: 1	PSOL–UNIÃO	17	far left: 1; left: 3; center-left: 4; center-right: 6; right: 9; far right: 3	PT–PP; 57.86%; 284 seats; overlap 6; J=0.27	MDB–UNIÃO; 49.98%; 265 seats; overlap 3; J=0.14
2022	2023.2	votes+seats; 63.79%; 350 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 2; far right: 2	PSOL–UNIÃO	15	far left: 1; left: 3; center-left: 4; center-right: 6; right: 9; far right: 3	PT–PP; 57.86%; 284 seats; overlap 7; J=0.30	MDB–UNIÃO; 49.98%; 265 seats; overlap 5; J=0.24
2022	2025.1	votes+seats; 54.45%; 291 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 2; far right: 1	PSOL–REPUBLICANOS	14	far left: 1; left: 3; center-left: 4; center-right: 6; right: 9; far right: 1	PT–PP; 57.86%; 284 seats; overlap 7; J=0.32	MDB–UNIÃO; 49.98%; 265 seats; overlap 4; J=0.19

The first result is that no observed cabinet coalition is itself a contiguous ideological interval. Every cabinet period has gaps inside its connected closure. The smallest number of gaps is eight, and several cabinet periods have more than fifteen. This means that Brazilian cabinet coalitions are not simply connected ideological blocs translated into government. They are selective coalitions: party sets assembled from a wider ideological span while omitting many parties that lie between their leftmost and rightmost members.

This matters for the interpretation of cabinet inversions. The observed cabinet inversions are sparse coalitions, but their connected closures are not inversions. All connected closures in Table 8 are vote-and-seat majorities. The 2016.2 cabinet, for example, has 46.54 percent of the vote and 259 seats, but its connected closure runs from PCdoB to PR and contains 20 parties. That closure has 77.61 percent of the vote and 408 seats. The inversion therefore does not come from a compact left-to-right ideological bloc. It comes from a selective cabinet coalition that retains a seat majority while omitting thirteen parties inside its ideological span.

The same logic applies to the more durable Temer-period inversion. The 2017.1 cabinet contains DEM, PMDB, PP, PPS, PSB, PSD, PSDB, and PV. Its connected closure runs from PSB to DEM, contains 25 parties, and has seventeen gaps. The observed cabinet has 49.91 percent of the vote and 266 seats, while the full PSB–DEM closure has 78.42 percent of the vote and 409 seats. The cabinet inversion is therefore not a connected ideological majority. It is a sparse center-to-right cabinet coalition located near the connected inversion region identified in the interval analysis. Its closest minimal connected winning interval is PSB–PTN, which is itself an inversion with 48.92 percent of the vote and 260 seats.

The Dilma coalitions have a different ideological profile from the later Temer coalitions. They are also sparse, but they are left-anchored. The cabinet sets before Temer include PT, PCdoB, and PDT while extending to parties such as PMDB, PSD, PR, PP, and PTB. Their closures run from PCdoB to either PR or PP, and their gap structure contains many center, center-right, and right parties that are not cabinet members. The impeachment transition changes this structure. From 2016.3 to 2016.4, PCdoB, PDT, PT, and PR leave, while PSB, PPS, PV, PSD, PSDB, PP, and DEM enter. In the ideology classification, the parties leaving are concentrated on the left and center-left, with one right party, while the parties entering are concentrated in the center, right, and far right. The cabinet remains punctured, but its ideological anchor shifts rightward.

The ideology diagnostics in the table confirm this transition. On the project's ideology scale, the average unweighted cabinet ideology rises from the pre-Temer sequence to the post-impeachment Temer sequence. The transition from 2016.3 to 2016.4 alone increases the unweighted cabinet ideology mean by 1.53 points and the seat-weighted mean by 1.49 points. This is the substantive difference between the two sparse coalition types. The pre-Temer coalitions are left-anchored and broad. The Temer coalitions are center-to-right and broad. Both are non-contiguous, but they occupy different regions of the ideological order.

The Bolsonaro-period cabinets provide a domain contrast. The PSC-inclusive 2021.3 and 2022.1 cabinets each have 47.25 percent of the vote and exactly 257 seats, so both observed coalitions are inversions. Their connected closure, PSD–DEM, has 57.37 percent of the vote and 292 seats and is a vote-and-seat majority. The closest minimal connected winning interval is PSDB–DEM, which has 51.52 percent of the vote and 257 seats. Thus, the 2018 mandate contains observed cabinet inversions but no ideological-interval inversion.

The 2023.1 Lula cabinet returns to a left-anchored structure, but it is not a compact left coalition. It contains PSOL, PCdoB, PT, PDT, PSB, REDE, MDB, PSD, and UNIÃO. Its connected closure runs from PSOL to UNIÃO, contains 26 parties, and has seventeen gaps. The observed cabinet is an inversion, with 48.89 percent of the vote and 263 seats, but its connected closure has 80.64 percent of the vote and 407 seats. The closest minimal connected winning interval is PT–PP, a broad vote-and-seat majority with 57.86 percent of the vote and 284 seats. The closest minimal connected inversion is MDB–UNIÃO, which lies to the right of the cabinet's left anchor.

This comparison is crucial for interpreting the 2022 result. The Lula cabinet is a realized inversion, but it is not the compact connected inversion revealed by the interval search. The compact connected inversion is PP–PL: an eight-party right/far-right interval with 45.35 percent of the vote and 258 seats. The observed Lula coalition reaches a seat majority through a broad and sparse cross-ideological cabinet arrangement. The connected interval analysis shows that the most compact vote-minority seat-majority bloc in the same Chamber is located on the right.

The two analyses therefore meet at a specific structural point. Cabinet coalitions identify realized governing party sets. Connected intervals identify the seat-vote geometry of ideological blocs. The cabinet-to-interval comparison shows that realized cabinets may draw on seat asymmetries without being connected blocs themselves. In 2014, the pre-Temer coalitions are sparse and left-anchored, while the Temer coalitions are sparse and shifted rightward. In 2018, two right-side cabinets reach exactly 257 seats without a vote majority, while their connected closures have both vote and seat majorities. In 2022, the Lula cabinet is a broad realized inversion, while the compact connected inversion lies on the right. The gap between cabinet coalitions and connected intervals is therefore not a nuisance. It shows how Brazil's coalition presidentialism combines sparse cabinet construction with an ideological distribution in which seat-efficient connected blocs are often located elsewhere.